

Bar Method Exercise Sheet

Part-Whole & Comparison Models

Part 1: Visualizing Addition

Example: Finding Total Pencils

Maya has **5 red pencils** and **3 blue pencils**. How many pencils does she have in total?

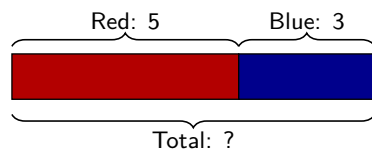
① **Identify and label the parts**

Determine the quantities given in the problem:

- Red pencils = 5
- Blue pencils = 3

② **Draw bars representing each part**

Draw connected blocks proportional to each value to visualize the whole:



③ **Write and solve the equation**

Combine the parts to calculate the overall total:

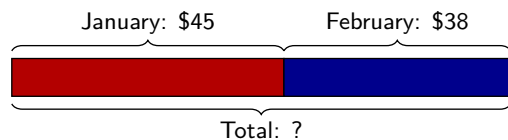
$$5 + 3 = 8 \quad \text{or} \quad 3 + 5 = 8$$

Maya has **8 pencils** altogether.

Practice Questions: Addition & Subtraction

Read each problem, fill in the missing numbers in the bar model, and solve.

1. Lucas saved **\$45 in January** and **\$38 in February**. How much money did he save in total?

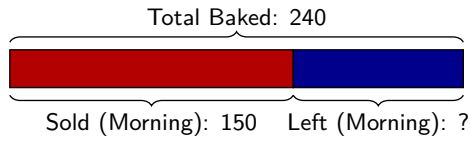


Equation: _____ + _____ = _____

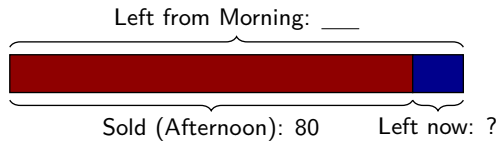
Total savings = \$ _____

2. John the baker made 240 cupcakes.

- (a) 150 cupcakes were sold in the morning. How many cupcakes were left?
 (b) If 80 cupcakes were sold in the afternoon, how many cupcakes does he have now?

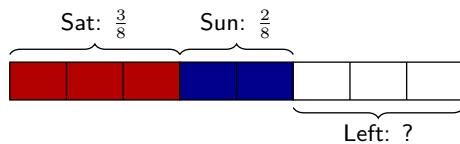


(a) Equation: _____ - _____ = _____ Cupcakes left = _____



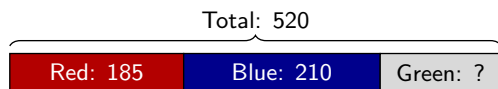
(b) Equation: _____ - _____ = _____ Cupcakes left = _____

3. Maya read $\frac{3}{8}$ of a book on Saturday and $\frac{2}{8}$ of it on Sunday. What fraction of the book does she have left to read?



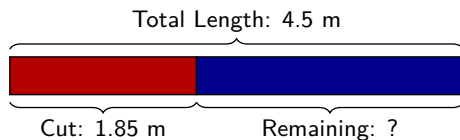
Fraction read = _____ + _____ = _____ Fraction left = $1 - \text{_____} = \text{_____}$

4. A toy store has 520 red, blue, and green balls. 185 are red and 210 are blue. How many green balls are there?



Green balls: $520 - 185 - 210 = \text{_____}$

5. A ribbon is 4.5 meters long. Hannah cuts off a piece measuring 1.85 meters. How long is the remaining ribbon?



Equation: _____ - _____ = _____ m Remaining ribbon = _____ m

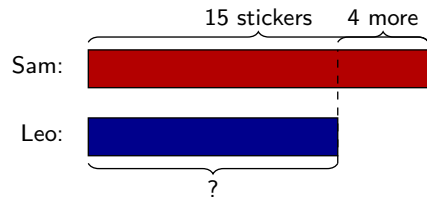
Part 2: Comparison Models (Two or More Quantities)

Example: Comparing Quantities

Sam has **15 stickers**. He has **4 more stickers** than Leo. How many stickers does Leo have?

① Draw parallel bars for each subject

Align the start of both bars to easily visualize the difference:



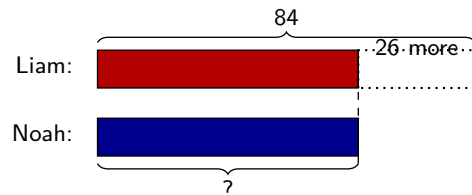
② Subtract the difference to find the smaller bar

$$15 - 4 = 11$$

Leo has **11 stickers**.

Practice Questions: Comparison Models

6. Liam has 84 stamps. He has 26 more stamps than Noah. How many stamps does Noah have?



Equation: _____ - _____ = _____

Noah's stamps = _____

7. Box A weighs 12 kg. Box B is 4 kg heavier than Box A. Box C is 3 kg lighter than Box B. What is the total weight of all three boxes?



Box B weight = _____ + _____ = _____ kg

Box C weight = _____ - _____ = _____ kg

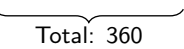
Total weight = _____ + _____ + _____ = _____ kg

8. Emma, Chloe, and Grace collected 360 seashells. Emma collected 40 more seashells than Chloe, and Grace collected twice as many seashells as Chloe. How many seashells did Chloe collect?

Chloe: 

Emma: 

Grace: 

 Total: 360

$$4 \text{ units} + 40 = 360 \Rightarrow 4 \text{ units} = \underline{\hspace{2cm}} \Rightarrow 1 \text{ unit} = \underline{\hspace{2cm}}$$

Chloe collected = $\underline{\hspace{2cm}}$ seashells

9. The ratio of the number of apples to oranges in a basket is 4 : 7. If there are 28 apples, how many oranges are there?

28 apples (4 units)
Apples: 

Oranges: 
? oranges (7 units)

$$1 \text{ unit} = 28 \div 4 = \underline{\hspace{2cm}}$$

$$\text{Oranges} = 7 \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

10. Ethan has 3 times as many marbles as Jacob. If Ethan has 36 more marbles than Jacob, how many marbles do they have altogether?

Difference: 36 marbles (2 units)
Ethan: 
Jacob: 

$$2 \text{ units} = 36 \Rightarrow 1 \text{ unit} = \underline{\hspace{2cm}}$$

$$\text{Total units} = 3 + 1 = 4 \text{ units} \Rightarrow \text{Altogether} = 4 \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ marbles}$$

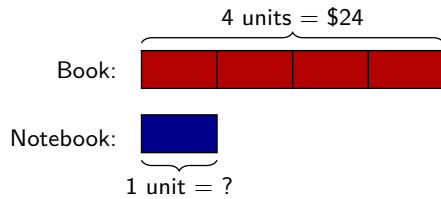
Part 3: Repeated Identity & Scaling (Multiplication & Division)

Example: Scaling with Equal Units

A book costs **4 times as much** as a notebook. If the book costs **\$24**, how much does the notebook cost?

① **Draw unit blocks for each item**

Represent the notebook as 1 unit and the book as 4 equal units:



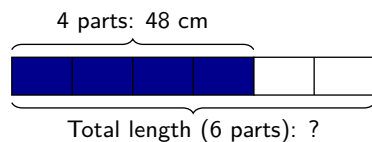
② **Divide total value by number of units**

$$4 \text{ units} = \$24 \implies 1 \text{ unit} = 24 \div 4 = 6$$

The notebook costs **\$6**.

Practice Questions: Scaling & Repeated Identity

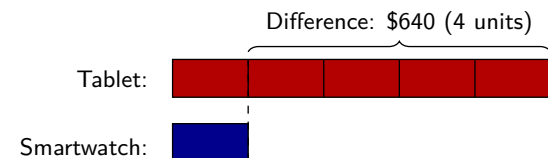
11. A length of rope is divided into 6 equal parts. If 4 parts measure 48 cm, what is the total length of the rope?



$$1 \text{ part} = 48 \div 4 = \underline{\hspace{2cm}} \text{ cm}$$

$$\text{Total length} = 6 \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ cm}$$

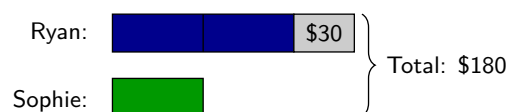
12. A tablet costs 5 times as much as a smartwatch. If the tablet costs \$640 more than the smartwatch, how much does the smartwatch cost?



$$4 \text{ units} = \$640 \implies 1 \text{ unit} = \$640 \div 4 = \underline{\hspace{2cm}}$$

$$\text{Smartwatch cost (1 unit)} = \$\underline{\hspace{2cm}}$$

13. Ryan and Sophie had \$180 in total. After Ryan spent \$30, he had 2 times as much money as Sophie. How much money did Ryan have initially?

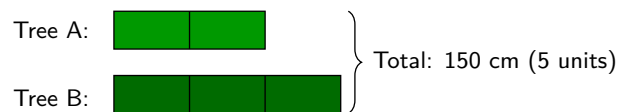


Remaining total = $\$180 - \$30 = \underline{\hspace{2cm}}$

3 units = $\underline{\hspace{2cm}}$ $\Rightarrow$ 1 unit = $\underline{\hspace{2cm}}$

Ryan initially = 2 units + \$30 = $2 \times \underline{\hspace{2cm}} + 30 = \$\underline{\hspace{2cm}}$

14. The total height of Tree A and Tree B is 150 cm. Tree A is $\frac{2}{3}$ the height of Tree B. Find the height of Tree B.



5 units = 150 cm $\Rightarrow$ 1 unit = $\underline{\hspace{2cm}}$ cm

Height of Tree B (3 units) = $3 \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$ cm

Part 4: Advanced / Multi-Step Problems

Example: Working Backwards / Constant Difference

Mark has **\$90** and Sam has **\$30**. How much money must Mark give to Sam so they both have the **same amount**?

① **Visualize the difference between the two quantities**

Calculate how much extra money Mark has compared to Sam:



② **Divide the difference into 2 equal parts**

To make both sides equal, split the extra difference in half and give 1 part to Sam:

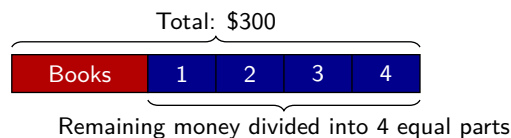
$$\text{Difference} = 90 - 30 = 60$$

$$\text{Amount to give} = 60 \div 2 = \mathbf{30}$$

Mark must give Sam **\$30**.

Practice Questions: Advanced Problems

15. David had \$300. He spent $\frac{1}{3}$ of his money on books and $\frac{1}{4}$ of the remaining money on lunch. How much money did he have left?

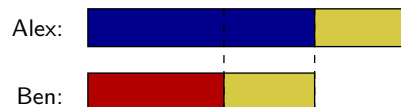


$$\text{Money after books} = \$300 - \frac{1}{3}(300) = \underline{\hspace{2cm}}$$

$$\text{Lunch cost} = \frac{1}{4} \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$\text{Money left} = \underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \$\underline{\hspace{2cm}}$$

16. Alex had \$120 and Ben had \$45. After both received the same allowance amount from their parents, Alex had twice as much money as Ben. How much allowance did each receive?



$$\text{Difference in money (unchanged)} = \$120 - \$45 = \$\underline{\hspace{2cm}}$$

$$\text{In the end, Alex has 2 units and Ben has 1 unit. Difference} = 1 \text{ unit} = \$\underline{\hspace{2cm}}$$

$$\text{Ben's final amount} = \$\underline{\hspace{2cm}} \Rightarrow \text{Allowance} = \underline{\hspace{2cm}} - 45 = \$\underline{\hspace{2cm}}$$

17. In a classroom, $\frac{3}{5}$ of the students were boys. After 6 boys left and 6 girls joined the class, $\frac{1}{2}$ of the students were boys. How many students were in the classroom altogether?

Before: Boys: 6/10 Girls: 4/10

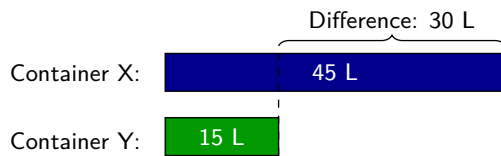
After: Boys: 5/10 Girls: 5/10

Before fraction of boys = $\frac{3}{5} = \frac{6}{10}$. After fraction of boys = $\frac{1}{2} = \frac{5}{10}$.

Change in boys = $\frac{6}{10} - \frac{5}{10} = \frac{1}{10}$ of total class.

$\frac{1}{10}$ of class = 6 students $\Rightarrow$ Total students = $6 \times 10 =$ _____

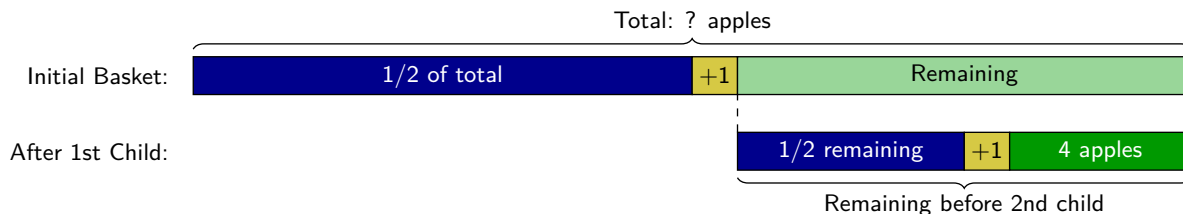
18. Container X has 45 liters of water and Container Y has 15 liters. How much water must be poured from Container X into Container Y so that both containers have equal volumes?



Difference = $45 - 15 = 30$ liters.

Amount to transfer = Difference $\div 2 = 30 \div 2 =$ _____ liters

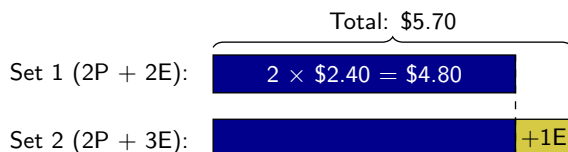
19. A basket of apples was shared among three children. The first child took $\frac{1}{2}$ of the apples plus 1 apple. The second child took $\frac{1}{2}$ of the remaining apples plus 1 apple. The third child received the remaining 4 apples. How many apples were in the basket at first?



Step 1 (Before 2nd child): $(4 + 1) \times 2 =$ _____ apples

Step 2 (Before 1st child / Total): $(\text{_____} + 1) \times 2 =$ _____ apples

20. A pencil and an eraser together cost \$2.40. Two pencils and three erasers together cost \$5.70. What is the cost of one pencil?



Cost of 1 eraser = $\$5.70 - \$4.80 = \$$ _____

Cost of 1 pencil = $\$2.40 - \$$ _____ = \$ _____